# Eco3121 Midterm Review

**Phoebe He**
**Economics Club**

# Outline of Midterm

- Single Linear Regression
  - Assumptions (6)
  - Derive OLS
  - Unbiasedness
  - Var of Estimators [Gauss-Markov]
  - Hypothesis test
- Multiple Linear Regression
  - Assumptions (6)
  - Omitted variable bias
  - Interactions & Dummy var

# Assumptions of SLR

Clanic Ass.

- **Assumption SLR.1 (Linear in parameters)**

$$y = \beta_0 + \beta_1 x + u$$

In the population, the relationship between y and x is linear

- **Assumption SLR.2 (Random sampling)**

$$\{(x_i, y_i) : \ i = 1, \dots n\}$$

The data is a random sample drawn from the population

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

Each data point therefore follows the population equation

- **Assumption SLR.3 (Sample variation in explanatory variable)**

$$\sum_{i=1}^{n} (x_i - \bar{x})^2 > 0$$

The values of the explanatory variables are not all the same (otherwise it would be impossible to study how different values of the explanatory variable lead to different values of the dependent variable)

- **Assumption SLR.4 (Zero conditional mean)**

$$E(u_i|x_i) = 0$$

The value of the explanatory variable must contain no information about the mean of the unobserved factors

+ SLR5: $Var(u|X_1 \dots X_n) = 0$

=> Variance of estimators

+ SLR6: Normality of estimators

=> Hypothesis test

# Derive OLS

→ ordinary Least square → $\hat{u}^2$

Population model: $Y = \beta_0 + \beta_1 X + u$

⇒ Estimated model: $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$ ⇒ $Y - \hat{Y} = \hat{u}$ → residual

⇒ Goal: $\min \sum \hat{u}_i^2 \Leftrightarrow \sum \hat{u}_i^2 = 0 \Rightarrow \sum (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2 = 0$.

By F.O.C of $\hat{\beta}_0$ & $\hat{\beta}_1$, we have

$$\begin{cases} \sum (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0 \quad (\sum \hat{u}_i = 0) \\ \sum X_i (Y - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0 \quad (\sum X_i \hat{u}_i = 0) \end{cases}$$

$$\Leftrightarrow \begin{cases} n\bar{Y} - n\hat{\beta}_0 - n\hat{\beta}_1 \bar{X} = 0 \\ n^2 \bar{X}\bar{Y} - n\bar{X}\hat{\beta}_0 - n\bar{X}\hat{\beta}_0 = 0 \end{cases} \Rightarrow \begin{cases} \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X} \\ \hat{\beta}_1 = \dfrac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sum (X_i - \bar{X})^2} = \dfrac{Cov(X,Y)}{Var(X)} \end{cases}$$

# Unbiasedness

What's Bias? $\quad Bias = E(\hat{\beta}_1) - \beta_1$

$\Rightarrow$ Goal: prove $E(\hat{\beta}_1) = \beta_1$

By OLS, $\hat{\beta}_1 = \dfrac{\sum(x_i-\bar{x})(y_i-\bar{y})}{\sum(x_i-\bar{x})^2}$

$$= \frac{\sum(x_i-\bar{x})}{\sum(x_i-\bar{x})^2}\left(\beta_0+\beta_1 x_i+u_i-(\beta_0+\beta_1\bar{x}+\bar{u})\right)$$

$$= \frac{\sum(x_i-\bar{x})}{\sum(x_i-\bar{x})^2}\left(\beta_1(x_i-\bar{x})+(u_i-\bar{u})\right)$$

$$= \beta_1 + \frac{\sum(x_i-\bar{x})(u_i-\bar{u})}{SST_x} \qquad \sum(x_i-\bar{x}) \to 0$$

$$\Rightarrow E(\hat{\beta}_1) = \beta_1 + E\left[\frac{\sum(x_i-\bar{x})(u_i-\bar{u})}{SST_x}\right] \qquad (u_i-\bar{u}) \to 0$$

$$= \beta_1 + \frac{E(\sum x_i u_i)}{SST_x} - \frac{E[\sum \bar{x} u_i]}{SST_x}$$

$= 0$

$\Rightarrow$ proved

# Var of Estimators

Recall SLR1 – SLR5 : $Var(u|X) = \sigma^2 \longrightarrow Var(y|X) = \sigma^2$

$$\hat{\beta_1} = \beta_1 + \frac{\Sigma(x_i - \bar{x})(u_i - \bar{u})}{\Sigma(x_i - \bar{x})^2}$$

Denote $w_i = \frac{x_i - \bar{x}}{SST_x}$ represent the weigh of each $u_i$

$$\Rightarrow \hat{\beta_1} = \beta_1 + \Sigma w_i u_i$$

$$\Rightarrow Var(\hat{\beta_1}) = Var(\Sigma w_i u_i) = \Sigma w_i^2 Var(u_i) = \sigma^2 \Sigma w_i^2$$

$$\because \Sigma w_i^2 = \left(\frac{\Sigma(x_i - \bar{x})}{SST_x}\right)^2 = \frac{SST_x}{SST_x^2} = \frac{1}{SST_x}$$

$$\Rightarrow Var(\hat{\beta_1}) = \frac{\sigma^2}{SST_x} \longrightarrow$$ To diminish $Var(\hat{\beta_1})$, we have Two Methods

$$Sd(\hat{\beta_1}) = \frac{\sigma}{\sqrt{SST_x}}$$

① Lower the deviation of error term ($\sigma^2$) → Numerator ↓.

② Increase the sample size ($\uparrow SST_x$) → Denominator ↑

Actually, we never know $\sigma$ for real.

⇒ Use an unbiased estimator $\hat{\sigma}$

We have $Var(u) = E(u^2) - \underbrace{E(u)^2}_{\text{SLR4}} = E(u^2) = \sigma^2$ (SLR4: $E(u) \to 0$)

⇒ find an unbiased estimator of $E(u^2)$

$$\hat{\sigma}^2 = \frac{\Sigma u_i^2}{n - k - 1} \Rightarrow df$$

($k$ → # of coefficients)

$$\Rightarrow Sd(\hat{\beta_1}) = \frac{\hat{\sigma}}{\sqrt{SST_x}} = \frac{\sqrt{\frac{SSR}{n-k-1}}}{\sqrt{SST_x}}$$

# Hypothesis test

SLR.1 – SLR.6 $\rightarrow$ $\begin{cases} u_i \sim N(0,1) \\ \hat{\beta}_1 = \beta_1 + \sum w_i u_i \end{cases} \Rightarrow \hat{\beta}_1 \sim N(\beta_1, se(\hat{\beta}_1))$.

Process: As we don't know actual $\sigma$. $\Rightarrow$ t-test

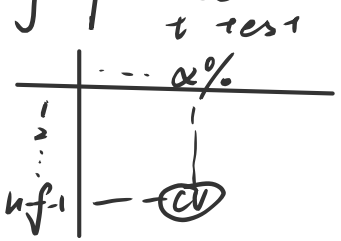
① Null / Alternative Hypothesis $\begin{cases} H_0: \beta_1 = 0 \\ H_1: \beta_1 \neq 0 \end{cases}$

② Compute test t-statistics of given $\hat{\beta}_1$.

$$t\text{-stat} = \frac{\hat{\beta}_1 - \beta_1}{se(\hat{\beta}_1)} \overset{\beta_1 \to 0}{\sim} t_{\alpha\%}(n-f-1) \xrightarrow{n \to \infty} \alpha f$$

$\alpha\%$ → stat significance

③ Find Critical Value: $t_{\alpha\%}(n-f-1)$




④ Compare CV with your t-stat

- t-stat > CV, Reject $H_0$
- t-stat < CV, Cannot reject $H_0$

# Assumptions of MLR

- **MLR.1 Linear in parameters.** The model in the population can be written as

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + u. \quad (2.1)$$

- **MLR.2 Random sampling.** We have a random sample of $n$ observations, $\{(x_{i1}, x_{i2}, \ldots, x_{ik}, y_i)_{i=1}^{n}\}$, drawn from the population model in MLR.1.
- **MLR.3 No perfect collinearity.** In the population and the sample, none of the independent variables is constant, and none is a *linear combination* of others.
- **MLR.4 Zero conditional mean.** In the population, $\mathrm{E}[u|x_1, x_2, \ldots, x_k] = 0$.

**MLR.5** The error $u$ has the same variance given any value of the explanatory variables.

$$\mathrm{Var}(u|\boldsymbol{x}_i) = \sigma^2.$$

The symbol $\boldsymbol{x}_i$ denotes all independent variables in the $i^{th}$ observation $\boldsymbol{x}_i = (x_{i1}, x_{i2}, \ldots, x_{ik})$

**MLR.6** The population error $u$ is *independent* of the explanatory variables $x_1, x_2, \ldots, x_k$ and is normally distributed with zero mean and variance $\sigma^2$: $u \sim \mathcal{N}(0, \sigma^2)$.

Similarly, MLR1-4 : unbiased
MLR1-5 : var of estimators
MLR1-6 : Hypothesis test

# Interpretation of MLR

$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \cdots + \hat{\beta}_n X_n$ => Interpret $\hat{\beta}_1$

Holding others constant, 1 unit increase in $x$ will lead to $\hat{\beta}_1$ units to $Y$.

① $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \cdots$ $\quad \partial y/\partial x_1 = \hat{\beta}_1 \Rightarrow \Delta y = \hat{\beta}_1 \Delta x$.

② $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 \ln x_1 + \cdots$ $\quad \partial y/\partial x_1 = \dfrac{\hat{\beta}_1}{x_1} \Rightarrow \Delta y = \hat{\beta}_1 \cdot \dfrac{\Delta x_1}{x_1} = \dfrac{\hat{\beta}_1}{100}\% \Delta x$.

③ $\ln \hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x + \cdots$ $\quad \partial y/\partial x = \partial y/\partial \ln y \cdot \dfrac{\partial \ln y}{\partial x} = y \cdot \hat{\beta}_1 \Rightarrow \dfrac{\Delta y}{y} = \hat{\beta}_1 \Delta x \Rightarrow \%\Delta y = 100 \hat{\beta}_1 \Delta x$

④ $\ln \hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x + \cdots$ $\quad \partial y/\partial x = \partial y/\partial \ln y \cdot \partial \ln y/\partial \ln x \cdot \dfrac{\partial \ln x}{\partial x} = y \cdot \hat{\beta}_1 \cdot \dfrac{1}{x}$

$\Rightarrow \dfrac{\Delta y}{y} = \hat{\beta}_1 \cdot \dfrac{\Delta x}{x} \Rightarrow \%\Delta y = \hat{\beta}_1 \%\Delta x$.

哪边有ln()，哪边有%.

# Unbiasedness & Var of estimators

1° Unbiased

Reg $x_1$ on $x_2 \cdots x_n$

$x_1 = \delta_0 + \delta_1 x_2 + \cdots + \delta_{n-1} x_n + r$

⇒ Similarly, $E(\hat{\beta}_1) = \beta_1 + E\left(\frac{\sum r_i u_i}{\sum r_i^2}\right) = \beta_1$

2° Var of estimators

$$Var(\hat{\beta}_j) = Var\left(\frac{\sum r_i u_i}{\sum r_i^2}\right) = \frac{\sum r_i^2 \, Var(u_i)}{(\sum r_i^2)^2}$$

$$= \frac{Var(u_i)}{\sum r_i^2} = \frac{\sigma^2}{SSR_j} = \frac{\sigma^2}{SST_j(1-R_j)}$$

⇒ Still, we never know $\hat{\sigma}$ for real

$$\hat{\sigma}^2 = \frac{SSR}{n-k-1} = \frac{SST(1-R^2)}{n-k-1}$$

- Ways to ↓ $Var(\hat{\beta}_j)$

① ↓ $\sigma^2$ (↓ noise)

② ↑ $SST_j$ (↑ sample)

③ ↓ $R_j$ ⇒ ↓ $x_j$ 与其它自变量 的 correlation

↳ Special case: Multicollineary

$R_j^2 \to 1 \Rightarrow Var(\hat{\beta}_j) \to \infty$

How to measure?

$$VIF = \frac{1}{1-R_j^2} \Rightarrow Var(\hat{\beta}_j) = \frac{\sigma^2}{SST_j} VIF$$

⇒ What if we put all variables with multicollineary into $u_i$?

→ omitted variable bias

# Omitted variable bias

the variables which can affect "y" & related to other $x_i$s are omitted.

- $Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$

  → $x_2 = \delta_0 + \delta_1 x_1 + v$ omitted

⇒ $Y = (\beta_0 + \beta_2\delta_0) + (\beta_2\delta_1 + \beta_1) x_1 + \beta_2 v + u$ ⇒ all biased

$E(\hat{\beta}_1) = \beta_2\delta_1 + \beta_1$

Direction of Bias of $\beta_1$

⇒ $\delta_1$

| | $Corr(\beta_1, \beta_2) > 0$ | $Corr(\beta_1, \beta_2) < 0$ |
|---|---|---|
| $\beta_2 > 0$ | + | − |
| $\beta_2 < 0$ | − | + |

# Hypothesis test

$H_0$ ① $\beta_i = \beta_j$ / $\beta_i - \beta_j = 0$ Single linear restriction $\Rightarrow t$

② $\beta_1 = 0, \beta_2 = 0 \cdots$ Muriple Exclusion restriction $\Rightarrow F$

① $H_0 : \beta_i = \beta_j$

# Method 1

Treat $(\beta_i - \beta_j)$ as a whole

$\Rightarrow$ t-stat of $(\beta_i - \beta_j)$ $\rightarrow$ Like SLR.

$$t = \frac{(\hat{\beta}_i - \hat{\beta}_j) - (\beta_i - \beta_j)}{se(\hat{\beta}_i - \hat{\beta}_j)} \rightarrow 0$$

$$\Rightarrow se(\hat{\beta}_i - \hat{\beta}_j) = \sqrt{Var(\hat{\beta}_i - \hat{\beta}_j)}$$

$$= \sqrt{Var(\hat{\beta}_i) + Var(\hat{\beta}_j) - 2Cov(\beta_i, \beta_j)}$$

May not be easy to compute

# Method 2

Recall $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + u$

To get $\theta = \beta_1 - \beta_2$. $Y = \beta_0 + \theta X_1 + \beta_2 (X_1 + X_2) + u$ ← New regressor

$\Rightarrow H_0 : \theta = 0$ → SLR.

② ★ $H_0: \beta_1 = 0,\ \beta_2 = 0,\ \beta_3 = 0$

For $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \varepsilon$.

$H_0 = \beta_1 \cdot \beta_2 \cdot \beta_3 = 0$.

$H_1$: $H_0$ is not true ( $\beta_1 \cdot \beta_2 \cdot \beta_3$ cannot be zero at the same time )

⇒ F-test

Unrestricted Model: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \varepsilon$.

Restricted Model: $y = \beta_0 + \beta_4 x_4 + \varepsilon$

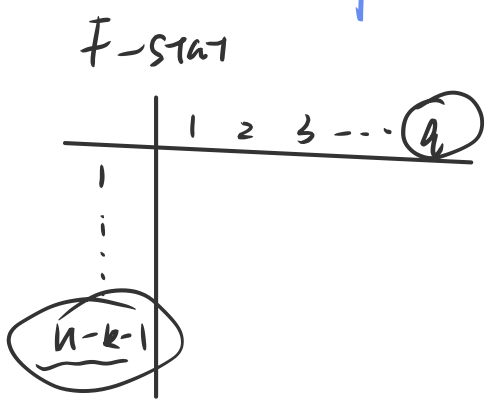
⇒ $F = \dfrac{(SSR_r - SSR_{ur})/q}{SSR_{ur}/n-k-1}$ → # of coefficient defference in ur & r.

→ df.

$\sim F_{q,\,n-k-1} = \dfrac{(R^2_{ur} - R^2_r)/q}{(1-R^2_r)/(n-k-1)}$

by $R^2 = 1 - \dfrac{SSR}{SST}$.

⇒ CU

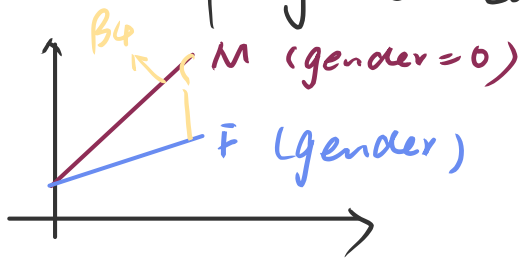


# Interaction & Dummy variable

$\log(wage) = \beta_0 + \beta_1 edu. + \beta_2 exp. + \beta_3 gender + \varepsilon.$

$\hookrightarrow \log(wage) = \beta_0 + \beta_1 edu. + \beta_2 exp. + \beta_3 gender + \beta_4 gender \cdot edu + \varepsilon.$

$\Rightarrow \partial \log(wage) / \partial edu = \beta_1 + \beta_4 \cdot gender \Rightarrow$




gender: Categrical / Dummy var. $\begin{cases} 0, \text{Male} \\ 1, \text{Famale} \end{cases}$

Adjusted $R^2$: $R^2$ will never ↓ as we add more vars. / Interpretable & precision.

recall $R^2 = \frac{SSE}{SST} = 1 - \frac{SSR}{SST}$

$\bar{R}^2 = 1 - \frac{n-1}{n-k-1} \times \frac{SSR}{SST}$ $\quad (k \uparrow \longrightarrow \frac{n-1}{n-k-1} \uparrow)$

$(n \uparrow \longrightarrow \frac{n-1}{n-k-1} \downarrow)$.